# Social media and search signals flag supply events in U.S. fresh produce markets

## Cultivate IQ – NAFSI Track 3 (Supply Chain)

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**Abstract**

Reddit talk and Google searches about a product's supply spike when something happens to that supply. Neither improved price forecasts in our tests. Reddit posts and comments on avocados, lettuce, onions and cucumbers from 2019 to 2026 were classified as price, supply or demand talk, and the weekly amounts were compared with USDA wholesale prices and shipments around 33 documented market events. Spikes in Reddit supply talk fell near these events twice as often as chance, and US Google searches for shortages and recalls spiked in the same weeks. Reddit tells what happened, such as an import halt, a crop disease or a recall; prices and shipments tell whether it mattered. Every unusual price rise came with a supply disruption: a border closure threat, a heat wave, a crop virus, a tariff. No food-safety outbreak, COVID-19 closure or viral recipe raised prices on its own. Adding Reddit, Google searches, news coverage, FDA recalls or Wikipedia pageviews to price history did not improve forecasts of national prices one or four weeks ahead, and around the events prices often moved before the talk did.

## 1. Introduction

Shoppers, grocery staff, cooks and growers write online about shortages, recalls and price jumps as they meet them. Online text has tracked such events before: search queries followed influenza (Ginsberg et al., 2009), and tweets followed food prices in Indonesia (Kim et al., 2017). In this report, Reddit talk about fresh produce is compared with USDA Market News wholesale prices and shipments.

Two questions are addressed: whether Reddit shows what lies behind a U.S. produce price move, and whether it forecasts prices. Avocados, lettuce (iceberg and romaine), dry onions and cucumbers are studied from January 2019 to September 2026. All four are perishable, priced every trading day in every terminal market and often named in conversation. The eight years cover the COVID-19 closures and enough documented events to compare event weeks with ordinary weeks and to test forecasts on data not used for training. The signal is tested on 33 documented market events chosen by fixed rules (Table 1, Appendix D).

Figure 1 shows two cases from 2022. When the United States halted Mexican avocado imports, Reddit carried the news the next day; shipments then fell and prices rose. When a virus cut California lettuce yields, a restaurant worker described the disease in August, but most talk came in November, when prices peaked. In both cases Reddit told what happened, and prices and shipments told how much it mattered. Sections 3.1–3.2 test this pattern on all 33 events, and Sections 3.3–3.4 test whether online talk forecasts prices.

## 2. Data and methods

**Data.** Reddit posts are public and timestamped. Posts from topical and city subreddits, the comments under product threads, and r/news headlines were collected through the Arctic Shift archive (Table 2). Bots, moderator notices, crossposts, paid and self-promotion, link farms, very short texts and duplicates were removed (Appendix A). The remaining texts form the event dataset. Each row is one text and one product it names, with the date, subreddit, city where known, and the event types its words match in the keyword dictionary (price change, supply shortage, demand change, weather, food-safety recall, promotion). A

**Table 1.** Documented events, January 2019 to September 2026, by kind. Each event has a product, dates and a news or agency source (Appendix D).

| Kind | Rule | Examples | Events |
|---|---|---|---|
| Food-safety outbreaks | Every CDC multistate outbreak linked to one of the products | Salinas romaine E. coli (2019), red onion Salmonella (2020), McDonald's onion E. coli (2024), iceberg Cyclospora (2026) | 10 |
| Supply disruptions | Reported in the produce trade press: import halts, border actions, tariffs, crop disease, weather | US halt of Mexican avocado imports (2022), INSV lettuce virus (2022), tariff on Mexican imports (2025), Florida freeze (2026) | 18 |
| Demand shocks | Reported nationally | COVID-19 closures (2020, one event per product), viral cucumber salad (2024) | 5 |

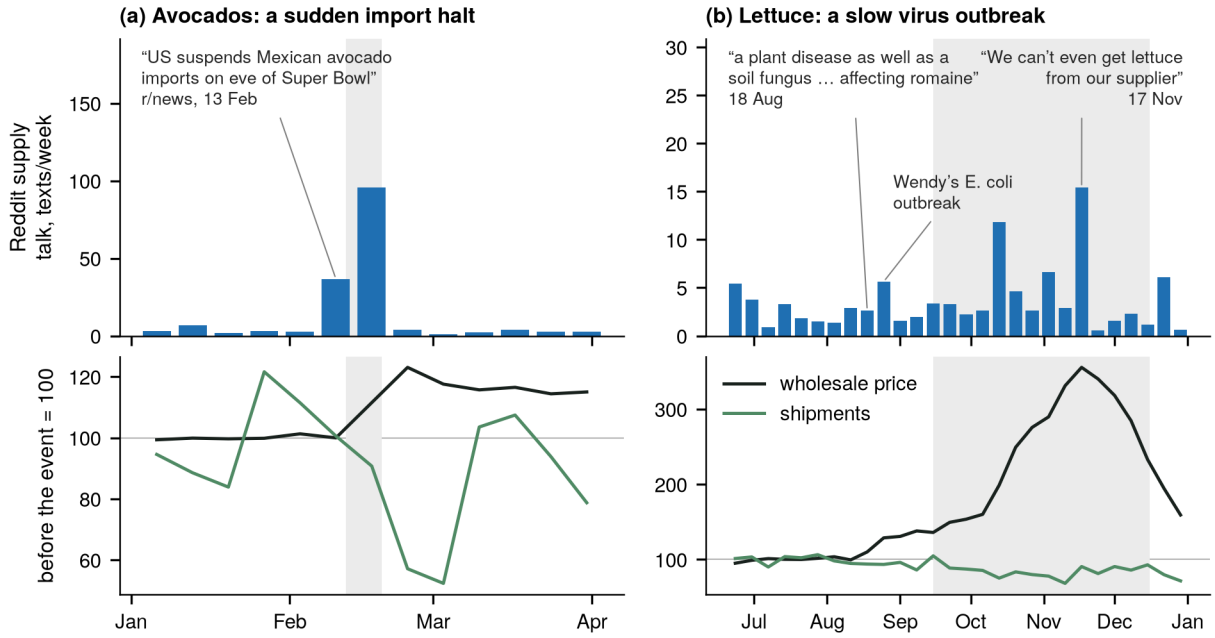



**Figure 1.** Two supply shocks up close. Top: weekly Reddit supply talk, with quotes from Reddit posts. Bottom: national wholesale price and shipments, each scaled to 100 before the event. Grey bands mark the events.

summary table counts the rows by product and event type (Appendix F). USDA AMS gives the market side: daily wholesale prices, combined into a weekly price index per product, and weekly shipments from US growing regions and imports. Prices and shipments together tell a supply shock from a demand shift. Google Trends gives a second online signal: weekly US searches for each product alone and with "price", "shortage" and "recall". Every source was collected from January 2019 to September 2026 in three periods with the same pipeline. The forecasting test adds three more free sources: US news coverage (GDELT), FDA food recalls (openFDA) and Wikipedia pageviews.

**Labels.** Each text gets one label for each product it names: price, supply, demand or other (Appendix B). Supply covers availability, shortages, crop problems, imports and recalls; avoiding a product after a scare counts as demand. One text per thread from 2022–2024 was labelled, most of them drawn with market keywords so that the rare labels are well represented. Two Claude models labelled the sample; 1,118 texts were also labelled independently by OpenAI Codex and ChatGPT. The 1,458 reference labels combine AI agreement with targeted team review: 388 have documented human review or adjudication, while the

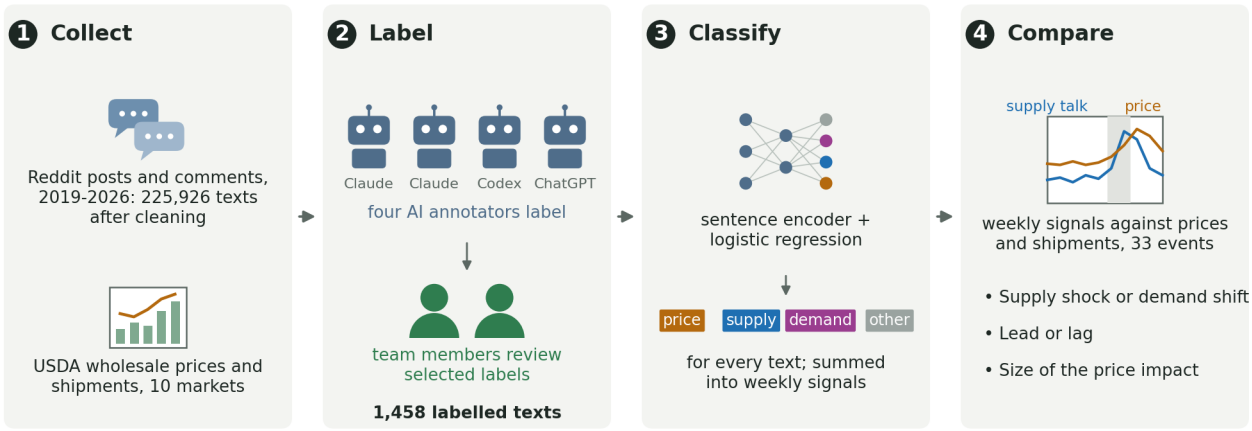



**Figure 2.** Study workflow. Four AI annotators label a sample of texts, and team members review selected labels. A classifier trained on these labels classifies every text, and its weekly signals are compared with USDA prices and shipments around documented events.

**Table 2.** Data sources, January 2019 to September 2026.

| Source | Data used | Coverage |
|---|---|---|
| Reddit, via the Arctic Shift archive | Posts, comments and r/news headlines that name a product | 29 topical subreddits (cooking, grocery, food service, growing, economy) and 9 city subreddits; 225,926 texts after cleaning |
| USDA AMS terminal market reports | Daily wholesale prices | 10 cities |
| USDA AMS movement reports | Weekly shipments from US growing regions and imports | National |
| Google Trends | Weekly search interest | US; 4 search terms per product |
| GDELT | Daily news volume and tone | US English-language news; 4 queries per product; forecasting only |
| openFDA | Food recall reports that name a product | US; forecasting only |
| Wikipedia | Daily pageviews of each product's article | English Wikipedia; forecasting only |
| Labelled sample | Texts labelled price, supply, demand or other | 1,458 texts from 2022–2024 |

remaining 1,070 have AI-agreement or ChatGPT-confirmed labels (Appendix B). In 100 adjudications, team members chose between two labels without knowing which model proposed which (Figure 2).

**Classifier.** Each text, preceded by the product name, is encoded with a pretrained sentence encoder (all-mpnet-base-v2) and labelled by a logistic regression trained on the labelled sample. It beats keyword rules and the zero-shot model of the challenge example, and matches a fine-tuned DistilRoBERTa that takes far longer to train (Table 3). The same model labels the texts of every year. Each week's amount of price, supply and demand talk is the sum of the predicted probabilities over that week's texts (Appendix C).

**Table 3.** Classifier accuracy in 5-fold cross-validation, grouped by Reddit thread.

| Model | Macro-F1 | Accuracy |
|---|---|---|
| Zero-shot BART-large-MNLI (challenge example) | 0.34 | 24% |
| Keyword rules (Task 1 dictionary) | 0.39 | 40% |
| Fine-tuned DistilRoBERTa | 0.71 | 75% |
| **all-mpnet + logistic regression (ours)** | **0.72** | **82%** |

## 3. Results

### 3.1 Supply talk spikes near real events

Spikes in supply talk point to real events. A week is counted as a spike when a series rises at least three standard deviations above its previous eight weeks. Supply-talk spikes fall within two weeks of a documented event twice as often as chance (Table 4). Spikes in price talk, demand talk, overall volume or sentiment are no more common near events than chance would give. Supply talk spiked around 20 of the 33 events, among them 9 of the 10 food-safety outbreaks, when people passed on recall notices (Figure 3).

**Table 4.** Spikes of at least three standard deviations above the previous eight weeks, January 2019 to September 2026, and how many lie within two weeks of a documented event. By chance, 21% of weeks would.

| Series | Spike weeks | Near an event | p |
|---|---|---|---|
| Reddit supply talk | 55 | 23 (42%) | 0.0003 |
| Reddit price talk | 47 | 14 (30%) | 0.10 |
| Reddit demand talk | 55 | 14 (25%) | 0.24 |
| Reddit overall volume | 54 | 13 (24%) | 0.33 |
| Reddit sentiment, either direction | 52 | 13 (25%) | 0.28 |
| Google searches, shortage or recall | 84 | 35 (42%) | < 0.0001 |

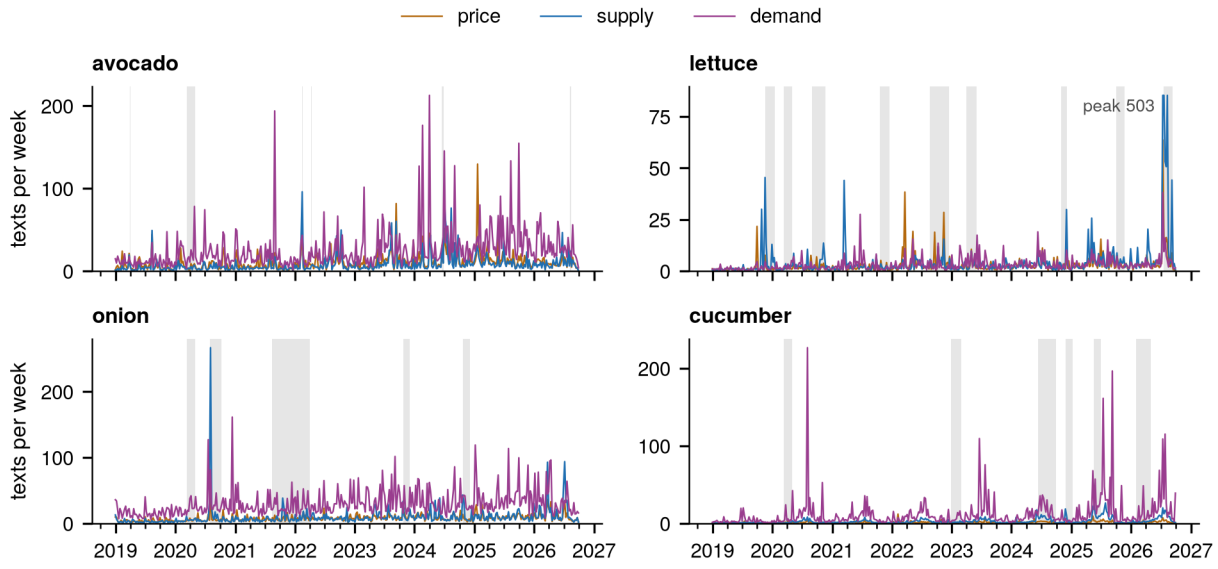



**Figure 3.** Weekly amount of price, supply and demand talk on Reddit per product (summed classifier probabilities). Grey bands mark the documented events. The lettuce axis is cut at the July 2026 Cyclospora outbreak.

US Google searches confirm the pattern. Searches for a product's shortage or recall spike near documented events as often as Reddit supply talk does (Table 4, Appendix E). Reddit or Google spiked around 24 of the 33 events, and in 13 of them the two peaked in the same week. Counting texts by their upvotes or comments finds no more events than counting each text once (Appendix G).

### 3.2 Prices show which events mattered

Every unusual price rise came with a supply disruption. A supply shock raises prices while shipments fall; a demand shift moves prices and shipments in the same direction. For each event, the event weeks and the two weeks after were compared with the eight weeks before, net of the same weeks two years before and after, and the change was ranked among all other windows of the same length (Table 5, Figure 4). Prices rose more than in nine of ten ordinary windows in eight events, all during supply disruptions. In six of them shipments fell as well. A carton of 48 Hass avocados went from $30 to $60 when the border closure was threatened in 2019, and a carton of 24 iceberg heads from $30 to $68 during the 2022 lettuce virus. No food-safety outbreak, COVID-19 closure or viral recipe raised prices on its own; the 2024 SunFed recall and the COVID-19 closures lowered cucumber prices.

**Table 5.** Events with an unusual price change: change in price and shipments over the event and the two weeks after, relative to the eight weeks before and beyond the seasonal norm. Supply talk: whether Reddit supply talk rose during the event.

| Event | Price | Shipments | Supply talk | Market reading |
|---|---|---|---|---|
| E15 Lettuce virus (INSV), 2022 | +121% | −13% | rose | **Supply shock** |
| E14 Wendy's romaine outbreak, 2022, start of E15 | +69% | −11% | – | **Supply shock** |
| E9 Northwest heat wave, onion crop, 2021–22 | +41% | −20% | rose | **Supply shock** |
| E1 US–Mexico border closure threat, avocado, 2019 | +40% | −12% | rose | **Supply shock** |
| E11 Onion Salmonella outbreak, 2021, inside E9 | +22% | −9% | rose | **Supply shock** |
| E26 Tariff on Mexican imports, cucumber, 2025 | +19% | −32% | – | **Supply shock** |
| E30 Salinas lettuce shortage, 2025 | +83% | 0% | – | Price rise, shipments level |
| E31 Florida freeze, cucumber, 2026 | +37% | +1% | – | Price rise, shipments level |
| E24 SunFed cucumber recall, 2024 | −25% | +3% | rose | Demand fall |
| E4 COVID-19 closures, cucumber, 2020 | −15% | −8% | – | Demand fall |

The other 21 events with a measured effect moved national prices no more than ordinary weeks do. Among them is the 2022 avocado import halt: it cut shipments by a quarter, and its price rise was within the ordinary range.

### 3.3 Online signals did not improve tested forecasts

None of the tested online sources improved national wholesale-price forecasts beyond price history. For each week from January 2021 to September 2026, the national price index one and four weeks ahead, and whether it would make a big move, were forecast with models trained only on earlier weeks. Each model used price history alone or price history plus one online source. The best forecasts use price history alone; no source improves them, and adding all sources together does not help either (Appendix H). Talk around the documented events shows the same (Section 3.4).

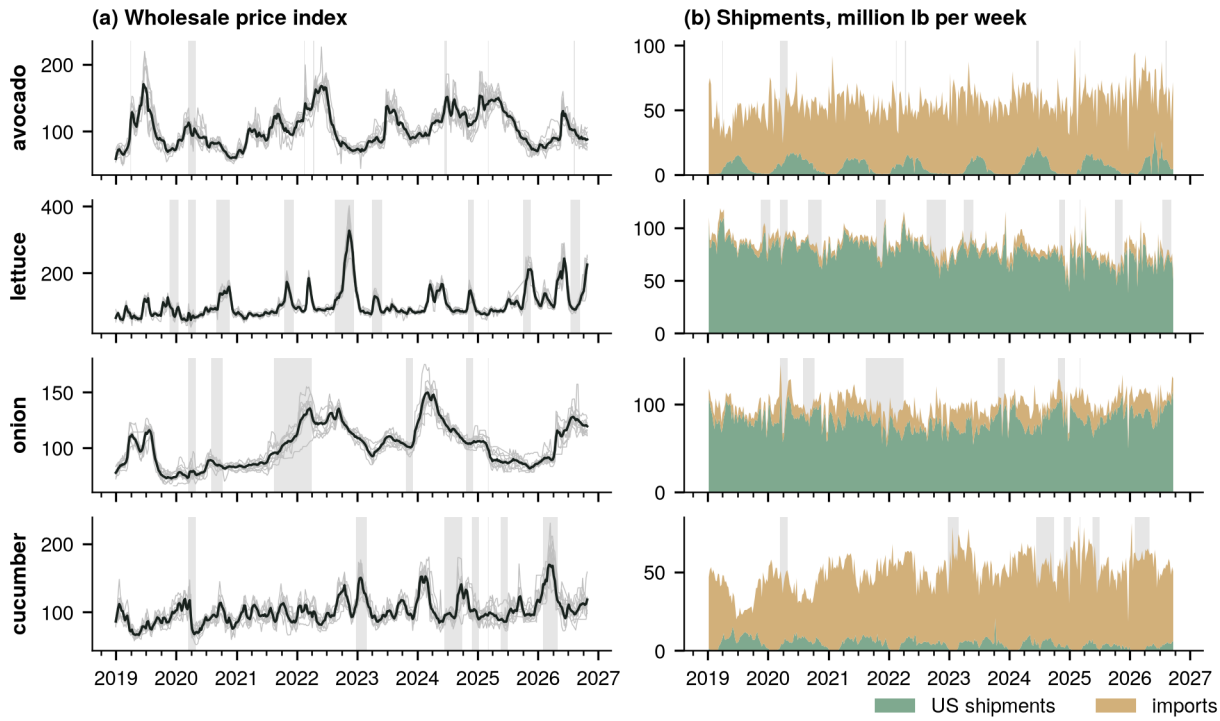



**Figure 4.** (a) Wholesale price index per product: all ten terminal markets pooled (black) and each market (grey), scaled to a 2019–2026 mean of about 100. (b) Weekly shipments from US growing regions and imports. Grey bands mark the documented events.

### 3.4 Talk does not lead prices

Reddit reports sudden shocks at once and explains slow ones as they peak. The avocado import halt was announced on Saturday, 12 February 2022. Supply talk surged the next day, mostly in r/news, and wholesale prices rose the following week (Figure 1a). The lettuce shortage built up over months. Prices began to climb in mid-August 2022, the week a restaurant worker wrote that a plant disease and a soil fungus were hitting lettuce in the Salinas growing region. The rise in lettuce talk later that month was about the Wendy's outbreak (Figure 1b). Most talk about the shortage itself came in mid-November, the week prices peaked: supplier outages, romaine at $5 a head, restaurant chains warning customers. Across all events, talk did not lead prices. In 27 events both moved. Prices moved more than a week before supply talk in 12, within a week of it in 8, and after it in 7; the median gap was one week, with prices first. Across all weeks, talk and prices correlate only weakly at any lag (Appendix E).

## 4. Uses of the signal

Supply-talk spikes point Market News reporters to the unusual weeks that have a story behind them. Produce buyers and food banks can see whether a shortage or a food-safety scare lies behind a price move, and food-safety agencies can see how far a recall travels. Google searches give a second check on the same week.

## 5. Limitations

**Who speaks.** Reddit users are younger and more urban than shoppers as a whole, and the choice of subreddits shapes what is seen. About one text in a hundred names a city, so the signal describes the national market.

**Events.** The trade press decides which supply disruptions are reported, and the dates of slow shortages are approximate. Two lettuce events in 2019 and 2020 have no comparison year free of other lettuce events,

so their price effect is not measured.

**Labels.** The classifier matches about four in five AI-assisted reference labels and makes most errors on demand, so demand talk is the noisiest signal. Only 388 of the 1,458 reference labels have documented human review or adjudication; the others were derived from AI agreement or confirmation. All labels come from 2022–2024, and texts from other years are classified by the same model.

**Market data.** Terminal market prices are wholesale prices for set packs and grades. Shipments count the pounds that Market News offices track. Google scales each search term on its own, and shortage and recall searches are zero in most weeks.

**Forecasting.** Only national wholesale prices of four products were forecast, not city or retail prices.

## Team science statement and use of AI tools

**Team.** The code was developed, the model designed, the data reviewed and the abstract and introduction written by Jiuyi Zheng. The data were prepared, and the conclusions and limitations written, by Gan Xu. Texts were labelled by Jiuyi Zheng, Gan Xu, Zhiguang Liu and Yizhen Wang; the data were analysed and the slides prepared by Yizhen Wang. The figures were made by Zhiguang Liu.

**AI tools.** Claude (Anthropic) wrote code under the team's direction, and Kiro, run with the challenge's AWS credits, reproduced the analysis from the README. Claude, Codex and ChatGPT labelled texts from the codebook; team members reviewed and adjudicated selected cases.

**Data and code.** All data, code and results are at github.com/joy91269/CultivateIQYummy; bash run_all.sh --analysis reruns the analysis in about ten minutes (deliverables in Appendix F).

## References

Ginsberg, J., Mohebbi, M. H., Patel, R. S., Brammer, L., Smolinski, M. S. and Brilliant, L. (2009). Detecting influenza epidemics using search engine query data. *Nature*, 457, 1012–1014.

Kim, J., Cha, M. and Lee, J. G. (2017). Nowcasting commodity prices using social media. *PeerJ Computer Science*, 3, e126.

# Appendix

**Table 6.** Where each challenge topic is covered.

| Challenge topic | Where |
|---|---|
| Data sources and the reasons for them | Section 2 (Data), Table 2 |
| Graphical workflow of the data collection | Figure 2 |
| Products and time period, and the reasons for them | Section 1 |
| Removal of bots, spam, duplicates and paid promotion | Section 2, Appendix A |
| Labelling of posts as price, supply, demand or other | Section 2 (Labels), Appendix B |
| Classification model and the reasons for it | Section 2 (Classifier) |
| Model accuracy | Table 3, Appendix C |
| Supply shock or demand shift for each major event, with evidence | Section 3.2, Table 5, Appendix D |
| Lead or lag of the online signal against the USDA price | Section 3.4, Appendix E |
| Price impact against a pre-event baseline | Section 3.2, Table 5, Appendix D |
| Predictive power of the online signal | Section 3.3, Appendix H |
| Users of the findings | Section 4 |
| Limitations | Section 5 |
| Team science and use of AI | Team science statement |

## A. Texts removed in cleaning

**Table 7.** Texts removed by each cleaning rule, January 2019 to September 2026.

| Rule | Removed |
|---|---|
| Same Reddit id downloaded twice | 1,047 |
| Outside the study period | 495 |
| Bots and bot messages | 512 |
| Moderator notices | 106 |
| Adult posts | 139 |
| Crossposts | 524 |
| Paid and self-promotion | 196 |
| Link farms (three or more links) | 1,071 |
| Under 15 characters | 2,425 |
| Same author, same text three or more times | 405 |
| Duplicate texts (first copy kept) | 2,368 |
| **Total** | **9,288** |

## B. Labels

The human labelling was split between two pairs of team members: Gan Xu and Jiuyi Zheng, and Zhiguang Liu and Yizhen Wang. Of the 1,458 gold reference labels, 388 have documented human review or adjudication: 150 challenging cases (100 assessed blind between candidate labels) and 238 additional checks. The other 1,070 gold labels reflect AI agreement or ChatGPT confirmation, not independent human annotation.

Most disagreements between the two team members are between demand and other. Full tables: outputs/tables/annotator_agreement.csv and human_agreement.csv.

**Table 8.** The four labels. The codebook (labeling/codebook.md) adds decision rules, for example price before supply before demand when a text has two signals, and other when in doubt.

| Label | The text is about… | Example |
|---|---|---|
| price | the product's price or a price change | "Avocados were $2.49 each at Kroger, insane." |
| supply | availability, shortages, crop or weather problems, imports, recalls | "No romaine anywhere, store says shortage." |
| demand | people buying or eating more or less of it: recipes centred on it, stocking up, trends, promotions, avoiding it after a scare | "Not touching onions until the E. coli thing is over." |
| other | no market signal: one ingredient among many, cooking tips, gardening, jokes | "How do you keep cut avocado from browning?" |

**Table 9.** Labelled texts by product and label.

| Product | Price | Supply | Demand | Other | Total |
|---|---|---|---|---|---|
| Avocado | 75 | 49 | 87 | 231 | 442 |
| Lettuce | 34 | 24 | 15 | 250 | 323 |
| Onion | 28 | 19 | 44 | 281 | 372 |
| Cucumber | 8 | 12 | 50 | 251 | 321 |
| All | 145 | 104 | 196 | 1,013 | 1,458 |

## C. Model details

The regularisation strength of the logistic regression is chosen inside each training fold, and the 95% interval comes from resampling threads.

## D. Events and their results in full

Events were chosen from CDC, FDA, USDA and trade-press reports. Food-safety windows run from the day the food was named or recalled to the day CDC declared the outbreak over, at most ten weeks. Sources for every event are in config/events_reference.csv. The fall 2026 Salinas lettuce shortage is left out because it runs past the end of the data.

**Table 13.** Documented events.

| ID | Product | Event | Dates | Kind |
|---|---|---|---|---|
| E1 | Avocado | US-Mexico border closure threat | 29 Mar – 4 Apr 2019 | Supply |
| E2 | Lettuce | E. coli O157:H7 outbreak linked to Salinas romaine | 20 Nov 2019 – 15 Jan 2020 | Food safety |
| E3 | Avocado | COVID-19 restaurant closures and panic buying | 13 Mar – 30 Apr 2020 | Demand |

**Table 10.** Agreement between annotators.

| Annotators compared | Texts | Agreement | Cohen's $\kappa$ |
|---|---|---|---|
| Claude Opus vs Claude Sonnet | 1,118 | 94.0% | 0.81 |
| Codex vs ChatGPT | 1,118 | 90.9% | 0.79 |
| Claude vs OpenAI (four pairs) | 1,118 | 86.9–89.5% | 0.66–0.73 |
| All four AI annotators (Fleiss' $\kappa$) | 1,118 | | 0.72 |
| First vs second team member | 80 hardest | 62.5% | 0.43 |

| ID | Product | Event | Dates | Kind |
|---|---|---|---|---|
| E4 | Cucumber | COVID-19 restaurant closures and panic buying | 13 Mar – 30 Apr 2020 | Demand |
| E5 | Lettuce | COVID-19 restaurant closures and panic buying | 13 Mar – 30 Apr 2020 | Demand |
| E6 | Onion | COVID-19 restaurant closures and panic buying | 13 Mar – 30 Apr 2020 | Demand |
| E7 | Onion | Salmonella Newport outbreak linked to red onions (Thomson International) | 31 Jul – 8 Oct 2020 | Food safety |
| E8 | Lettuce | INSV and soil-disease lettuce shortage in Salinas | 1 Sep – 20 Nov 2020 | Supply |
| E9 | Onion | Pacific Northwest heat wave cuts onion crop | 15 Aug 2021 – 31 Mar 2022 | Supply |
| E10 | Lettuce | Fall 2021 lettuce shortage at Salinas-Yuma transition | 18 Oct – 15 Dec 2021 | Supply |
| E11 | Onion | Salmonella Oranienburg outbreak linked to onions from Chihuahua (ProSource) | 20 Oct – 29 Dec 2021 | Food safety |
| E12 | Avocado | US suspends Mexican avocado imports | 12 Feb – 18 Feb 2022 | Supply |
| E13 | Avocado | Texas enhanced border truck inspections | 8 Apr – 15 Apr 2022 | Supply |
| E14 | Lettuce | Wendy's romaine E. coli outbreak | 19 Aug – 4 Oct 2022 | Food safety |
| E15 | Lettuce | INSV lettuce shortage in California | 15 Sep – 15 Dec 2022 | Supply |
| E16 | Cucumber | Winter 2022-23 cucumber shortage | 26 Dec 2022 – 28 Feb 2023 | Supply |
| E17 | Lettuce | Spring 2023 lettuce shortage after Salinas flooding | 1 Apr – 31 May 2023 | Supply |
| E18 | Onion | Salmonella Thompson outbreak linked to Gills Onions diced onions | 23 Oct – 4 Dec 2023 | Food safety |
| E19 | Cucumber | Salmonella outbreak linked to cucumbers | 12 Jun – 21 Aug 2024 | Food safety |
| E20 | Avocado | USDA pauses avocado inspections in Michoacan | 14 Jun – 28 Jun 2024 | Supply |
| E21 | Cucumber | Viral cucumber salad trend | 15 Jul – 30 Sep 2024 | Demand |
| E22 | Onion | McDonald's E. coli outbreak traced to slivered onions | 22 Oct – 3 Dec 2024 | Food safety |
| E23 | Lettuce | Fall 2024 lettuce shortage at Salinas-desert transition | 1 Nov – 6 Dec 2024 | Supply |
| E24 | Cucumber | SunFed cucumber recall (Salmonella Typhimurium) | 27 Nov 2024 – 8 Jan 2025 | Food safety |
| E25 | Avocado | 25% US tariff on Mexican imports | 4 Mar – 6 Mar 2025 | Supply |
| E26 | Cucumber | 25% US tariff on Mexican and Canadian imports | 4 Mar – 6 Mar 2025 | Supply |
| E27 | Lettuce | 25% US tariff on Mexican and Canadian imports | 4 Mar – 6 Mar 2025 | Supply |
| E28 | Onion | 25% US tariff on Mexican and Canadian imports | 4 Mar – 6 Mar 2025 | Supply |
| E29 | Cucumber | Salmonella Montevideo outbreak linked to Bedner Growers cucumbers | 19 May – 30 Jun 2025 | Food safety |
| E30 | Lettuce | Fall 2025 Salinas lettuce shortage | 1 Oct – 20 Nov 2025 | Supply |
| E31 | Cucumber | Florida freeze cucumber shortage | 1 Feb – 30 Apr 2026 | Supply |
| E32 | Lettuce | Cyclospora outbreak linked to Taylor Farms de Mexico iceberg lettuce | 17 Jul – 11 Sep 2026 | Food safety |
| E33 | Avocado | US suspends avocado inspections in Michoacan | 5 Aug – 13 Aug 2026 | Supply |

**Table 14.** Each event in full. Rank: percentile of the event's seasonally adjusted change among all other windows of the same length that overlap no documented event of the product; – when no comparison year is free of other events. Typical price: median wholesale price of the most traded package before and during the event (Hass avocados, two-layer carton of 48; iceberg lettuce, carton of 24; cucumbers, 1 1/9 bushel carton; yellow onions, 50 lb sack). Talk: weekly rate during the event over the eight weeks before, with 95% interval.

| Event | Price rank | Shipments rank | Typical price, before → during | Supply talk | Demand talk |
|---|---|---|---|---|---|
| E1 avocado | 96 | 10 | \$30 → \$60 | 4.8 (2.7–8.4) | 1.3 (0.9–2.0) |
| E2 lettuce | – | – | \$25 → \$25.5 | 2.1 (1.4–3.1) | 0.9 (0.5–2.0) |
| E3 avocado | 14 | 23 | \$47.1 → \$49.6 | 0.6 (0.4–0.9) | 0.8 (0.7–1.0) |
| E4 cucumber | 8 | 17 | \$28.2 → \$19 | 3.8 (0.7–20) | 1.9 (1.2–3.1) |
| E5 lettuce | 51 | 0 | \$17.5 → \$19 | 1.4 (0.7–3.1) | 1.5 (0.7–3.1) |

| Event | Price rank | Shipments rank | Typical price, before → during | Supply talk | Demand talk |
|---|---|---|---|---|---|
| E6 onion | 13 | 54 | \$15.6 → \$14 | 2.0 (1.2–3.4) | 1.4 (1.2–1.8) |
| E7 onion | 79 | 69 | \$16.5 → \$16.5 | 8.9 (6.2–13) | 0.9 (0.8–1.0) |
| E8 lettuce | – | – | \$21 → \$33 | 1.6 (0.9–2.9) | 0.7 (0.4–1.4) |
| E9 onion | 100 | 0 | \$18.5 → \$25 | 1.4 (1.0–1.9) | 1.0 (0.8–1.1) |
| E10 lettuce | 76 | 16 | \$28 → \$34 | 0.7 (0.3–1.3) | 0.3 (0.2–0.7) |
| E11 onion | 93 | 4 | \$21 → \$25 | 2.2 (1.5–3.1) | 1.0 (0.8–1.2) |
| E12 avocado | 81 | 1 | \$58.5 → \$72 | 18 (12–26) | 3.8 (2.8–5.1) |
| E13 avocado | 64 | 68 | \$67.5 → \$81.5 | 0.2 (0.1–0.5) | 1.1 (0.8–1.5) |
| E14 lettuce | 94 | 11 | \$26 → \$39 | 1.1 (0.6–2.0) | 0.6 (0.3–1.1) |
| E15 lettuce | 100 | 15 | \$30 → \$68 | 1.8 (1.1–2.9) | 0.9 (0.5–1.6) |
| E16 cucumber | 45 | 57 | \$25.5 → \$29 | 1.1 (0.5–2.6) | 1.2 (0.8–1.7) |
| E17 lettuce | 74 | 73 | \$23 → \$31.5 | 2.3 (1.2–4.5) | 1.2 (0.8–1.8) |
| E18 onion | 67 | 42 | \$19 → \$17.5 | 1.8 (1.3–2.5) | 0.7 (0.6–0.8) |
| E19 cucumber | 50 | 80 | \$21.2 → \$20 | 1.8 (1.2–2.7) | 1.9 (1.4–2.4) |
| E20 avocado | 79 | 64 | \$55 → \$71 | 2.4 (1.6–3.6) | 1.7 (1.3–2.2) |
| E21 cucumber | 68 | 72 | \$20 → \$22 | 0.7 (0.5–0.9) | 0.8 (0.6–0.9) |
| E22 onion | 61 | 16 | \$19 → \$18 | 1.6 (1.2–2.2) | 0.8 (0.6–0.9) |
| E23 lettuce | 86 | 10 | \$22.5 → \$36.2 | 4.0 (2.2–7.5) | 1.2 (0.7–2.3) |
| E24 cucumber | 3 | 58 | \$23.5 → \$21 | 5.9 (2.5–14) | 0.5 (0.3–0.8) |
| E25 avocado | 26 | 18 | \$79.5 → \$80 | 0.5 (0.2–1.0) | 0.5 (0.3–0.8) |
| E26 cucumber | 95 | 0 | \$21 → \$25 | 1.0 (0.1–7.7) | 0.5 (0.2–1.3) |
| E27 lettuce | 42 | 33 | \$21.5 → \$21 | 1.3 (0.4–4.9) | 0.5 (0.1–2.4) |
| E28 onion | 11 | 47 | \$18.5 → \$16.5 | 0.4 (0.1–0.9) | 0.4 (0.3–0.7) |
| E29 cucumber | 26 | 55 | \$29 → \$24 | 2.1 (1.5–3.1) | 1.5 (1.2–1.8) |
| E30 lettuce | 97 | 55 | \$29.5 → \$50 | 0.8 (0.5–1.2) | 0.5 (0.3–0.8) |
| E31 cucumber | 100 | 56 | \$26 → \$38.5 | 1.7 (0.8–3.5) | 2.3 (1.6–3.3) |
| E32 lettuce | 11 | 43 | \$47.5 → \$28.8 | 23 (16–33) | 2.6 (1.8–3.9) |
| E33 avocado | 62 | 38 | \$49 → \$47.5 | 0.9 (0.6–1.3) | 0.9 (0.7–1.2) |

Recalls and products pulled from shelves count as supply talk in the codebook.

## E. Methods

**Price impact.** For each event, the mean of the log weekly price index over the event and the two weeks after is compared with its mean over the eight weeks before, and the same change is subtracted over the same calendar weeks two years before and after, skipping windows that overlap another event of the product; shipments are treated the same way. To judge the size of a change, the same statistic is computed for every other window of the same length in 2019–2026 that overlaps no documented event of the product, and the event is ranked among them (Table 14).

**Market reading.** A price rank of at least 90 with a shipments rank of at most 25 is a supply shock. An unusual price rise with flat or rising shipments, or an unusual fall with flat or falling shipments, is a demand shift. A price rank between 10 and 90 is no clear price effect.

**Online reading.** Supply and demand talk during the event (summed probabilities) is compared with the eight weeks before as a ratio of weekly rates with a 95% Poisson interval. A kind of talk rose when its interval lies above 1 and it rose more than the other talk about the product.

**Spike test.** Each weekly series, from Reddit or Google, is turned into a z-score against the previous eight weeks; a week spikes in Google shortage or recall searches when either term does. Of all product-weeks, 21%

**Table 11.** Classifier accuracy in detail.

| Model | Accuracy | Macro-F1 (95% CI) | Cohen's $\kappa$ |
|---|---|---|---|
| Keyword rules | 0.40 | 0.39 (0.37–0.42) | 0.20 |
| Zero-shot BART-large-MNLI | 0.24 | 0.34 (0.32–0.37) | 0.12 |
| Fine-tuned DistilRoBERTa | 0.75 | 0.71 (0.68–0.73) | 0.56 |
| **all-mpnet + logistic regression (ours)** | **0.82** | **0.72 (0.69–0.75)** | **0.61** |

**Table 12.** Per-label scores of the classifier. The last column sums the predicted probabilities over the labelled texts.

| Label | Precision | Recall | F1 | Labelled texts | Sum of probabilities |
|---|---|---|---|---|---|
| Price | 0.79 | 0.66 | 0.72 | 145 | 141 |
| Supply | 0.74 | 0.62 | 0.67 | 104 | 101 |
| Demand | 0.69 | 0.54 | 0.60 | 196 | 192 |
| Other | 0.85 | 0.92 | 0.89 | 1,013 | 1,023 |

lie within two weeks of a documented event. For the spikes of each series, the number that lie that close to an event is counted and its share tested against 21% with a one-sided binomial test (Table 4).

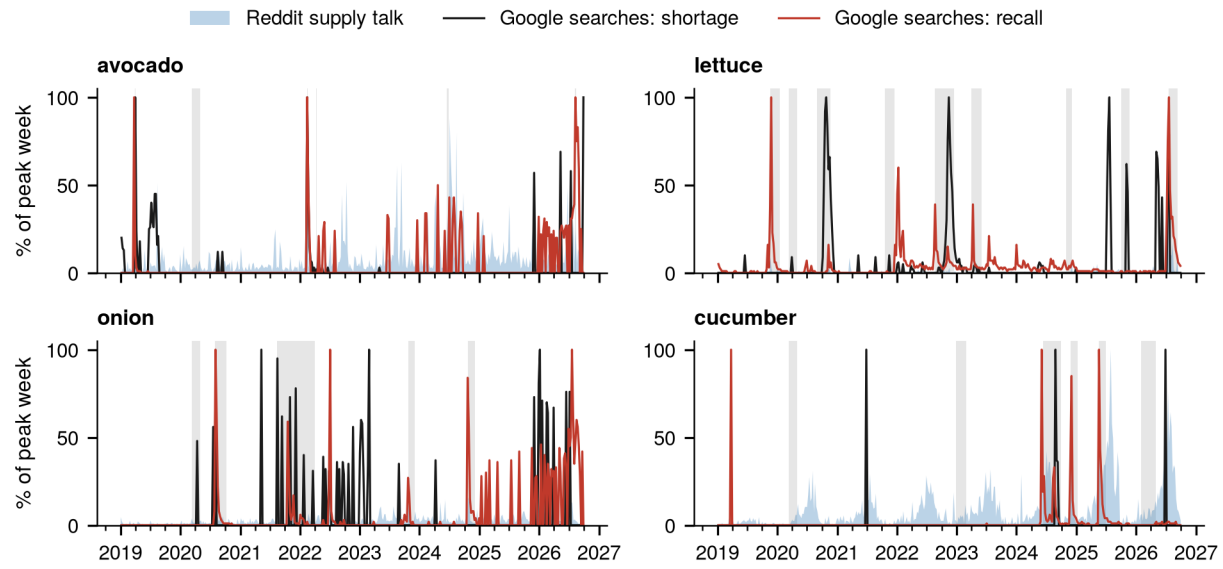



**Figure 5.** Reddit supply talk and US Google searches for each product's shortage and recall, each as a share of its busiest week within each collection period. Grey bands mark the documented events.

**Lead and lag.** For each product and series, the weekly z-score of the signal is correlated with that of the price index shifted by up to eight weeks either way. Because the series are autocorrelated and the best of 17 lags is chosen, the best correlation is tested by rotating the signal against the price by every offset of at least 13 weeks. All correlations are weak ($|r| \leq 0.22$). Five of the 20 product–series pairs have $p < 0.05$, three of them in the same week and two at a lag of seven weeks.

## F. Deliverables

**Table 15.** Challenge deliverables in the repository.

| Deliverable | File |
|---|---|
| Event dataset | data/processed/event_dataset.csv (2022–2024); event_dataset.csv.gz in data/processed_2019_2021/ and data/processed_2025_2026/ |
| Product and event keyword list | data/processed/keyword_dictionary.csv |
| Summary table | summary_table.csv in each of the three data/processed* folders |
| Dataset generation code | data_collection/ |
| Task 2 notebook | notebooks/task2_signals_and_prices.ipynb |
| Environment | requirements.txt |
| Data dictionary and provenance | docs/data_dictionary.md |
| Labels and codebook | labeling/labels.csv, labeling/codebook.md |

## G. Weighting texts by upvotes or comments

Each text's supply probability is multiplied by a weight from its upvotes or, for posts, its number of comments, and the spike test of Table 4 is rerun on the weighted supply series. To pick a weighting without testing it on the same data, it is chosen on 2019–2022 and tested on 2023–2026. The Google column does not use the list of events: it counts Reddit supply spikes with a Google shortage or recall spike within one week. No weighting finds more events than equal weights. The one chosen on 2019–2022, both: 1 + log(1 + upvotes + comments), finds 10 of 26 spikes near an event in 2023–2026, against 11 of 29 with equal weights. Notebook section 13; `weighting_robustness.csv`.

**Table 16.** Reddit supply spikes under different weights. Near an event: within two weeks of a documented event. Columns 2019–22 and 2023–26: spikes near an event out of all spikes in that period. With Google: spikes with a Google shortage or recall spike within one week.

| Weight per text | Spikes | Near an event (p) | 2019–22 | 2023–26 | With Google |
|---|---|---|---|---|---|
| Equal (main results) | 55 | 23 (42%, < 0.001) | 12/26 | 11/29 | 16 (29%) |
| Upvotes: 1 + log(1 + upvotes) | 39 | 16 (41%, 0.003) | 7/16 | 9/23 | 13 (33%) |
| Upvotes: $\times 2$ if $\geq 10$ | 50 | 20 (40%, 0.002) | 9/22 | 11/28 | 16 (32%) |
| Upvotes: $\times 3$ if $\geq 10$ | 49 | 18 (37%, 0.008) | 7/23 | 11/26 | 14 (29%) |
| Upvotes: raw count | 50 | 18 (36%, 0.010) | 10/28 | 8/22 | 17 (34%) |
| Comments: 1 + log(1 + comments) | 41 | 16 (39%, 0.006) | 7/17 | 9/24 | 16 (39%) |
| Comments: $\times 2$ if $\geq 10$ | 52 | 20 (38%, 0.003) | 9/23 | 11/29 | 16 (31%) |
| Comments: $\times 3$ if $\geq 10$ | 51 | 19 (37%, 0.005) | 8/24 | 11/27 | 18 (35%) |
| Both: 1 + log(1 + upvotes + comments) | 42 | 18 (43%, 0.001) | 8/16 | 10/26 | 14 (33%) |

## H. Forecasting prices, 2019–2026

The forecasting test uses the data of Table 2 and adds daily US news volume and tone from GDELT, FDA food recalls from openFDA and Wikipedia pageviews.

The targets are the change in the national log price index one and four weeks ahead, and whether the four-week change is a big move (at least the product's 85th percentile in the training weeks). The test runs from January 2021 to September 2026 in four-week blocks; each block uses models trained only on weeks

whose outcome was already known. Every feature set contains price history: recent weekly changes, distance from the 8-, 26- and 52-week means, change over a year, season and product. The other sets add one source as z-scores against the previous eight weeks. Ridge regression, gradient boosting and random forest were compared for changes, and logistic regression and gradient boosting for big moves. A block bootstrap over test weeks gives the p-values and intervals. Code and data: `forecast_check/`; tables `forecast_long_*.csv`.

**Table 17.** Forecasting the national wholesale price index, January 2021 to September 2026 (about 1,190 product-weeks). $R^2$ against a forecast of no change; AUC for big moves four weeks ahead. Each column uses the model that did best with price history alone: ridge regression (1 week), gradient boosting (4 weeks), logistic regression (AUC). With every model, no set is significantly better than price history alone (all p $\geq$ 0.07); no AUC gain exceeds 0.014, and every 95% interval includes zero.

| Features | $R^2$, 1 week | $R^2$, 4 weeks | AUC, big move |
|---|---|---|---|
| **Price history only** | **0.33** | **0.23** | **0.62** |
| + Shipments | 0.33 | 0.18 | 0.61 |
| + Reddit | 0.31 | 0.20 | 0.62 |
| + Google Trends | 0.33 | 0.17 | 0.61 |
| + News (GDELT) | 0.32 | 0.22 | 0.61 |
| + FDA recalls | 0.33 | 0.22 | 0.61 |
| + Wikipedia | 0.33 | 0.22 | 0.61 |
| + All online sources | 0.29 | 0.17 | 0.59 |